



Question Paper



MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech IVth Semester Midterm Examination March 2025

DATABASE SYSTEMS [CSE 2224]

Marks: 30

Duration: 90 mins.

MCQ

Answer all the questions.

Section Duration: 20 mins

- 1) What is data redundancy in a database system?

[The unnecessary duplication of data within a database](#)

[The process of normalizing data to remove duplicate records](#)

[The intentional replication of data for backup purposes](#)

[The practice of encrypting data for security reasons](#)

(0.5)
- 2) Which of the following statements best describes the difference between Procedural DML and Declarative DML?

[Procedural DML tells the database how to get the data step by step, while Declarative DML only tells what data is needed.](#)

[Procedural DML specifies what data is needed, while Declarative DML specifies how to retrieve it.](#)

[Procedural DML is used only in relational databases, while Declarative DML is used in non-relational databases.](#)

[There is no difference between Procedural DML and Declarative DML; both work the same way.](#)

(0.5)
- 3) Which type of database user is responsible for defining the structure of the database and managing its overall design?

[End Users](#)

[Database Administrator \(DBA\)](#)

[Application Programmers](#)

[Naïve Users](#)

(0.5)
- 4) What is the primary purpose of a data dictionary in a database system?

[To store actual data records in tables](#)

[To back up the entire database automatically.](#)

[To maintain metadata, such as table structures and constraints](#)

[To encrypt sensitive data for security purposes](#)

(0.5)
- 5) _____ is not a basic operator in relational algebra

[set difference: -](#)

[rename: ρ](#)

[Cartesian product: x](#)

[Set intersection : ∩](#)

(0.5)
- 6) Consider the entity relationship which consists of two entity sets, customer and loan, related through a binary relationship set borrower. An undirected line from the relationship set borrower to the entity set loan specifies that borrower is either a _____ or _____ relationship set from customer to loan.

[Many to many, one to many](#)

[One to one, many to one](#)

[Many to many, one to one](#)

[One to One, Many to Many](#)

(0.5)
- 7) Match the following

[Many to many, one to many](#)

[One to one, many to one](#)

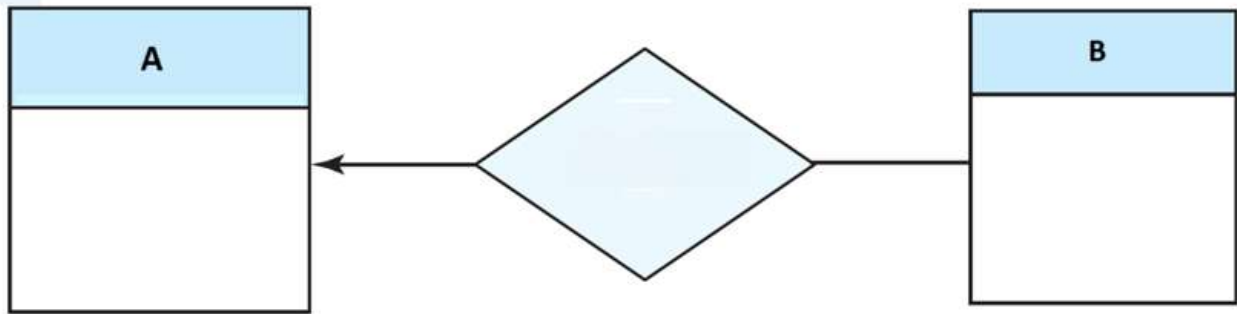
[Many to many, one to one](#)

[One to One, Many to Many](#)

(0.5)

Symbol	Name
i. σ	v. Cartesian Product
ii. π	vi. Natural Join
iii. $\bowtie$	vii. Projection
iv. $\times$	viii. Selection

8)



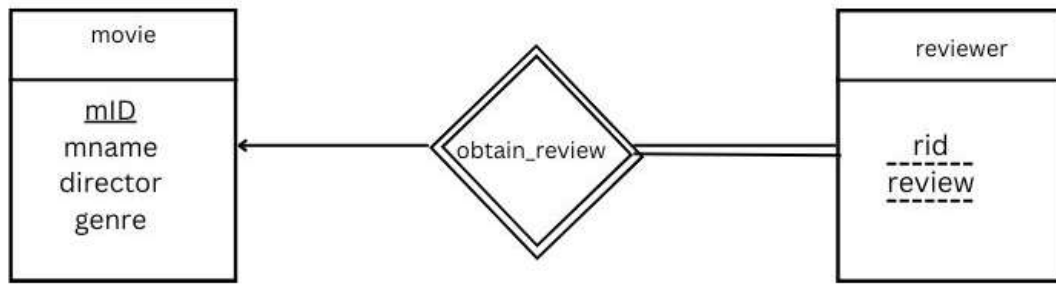
(0.5)

What type of relationship exists between the A and B entities in the given E-R diagram?

[One-to-One \(1:1\)](#), [One-to-Many \(1:M\)](#), [Many-to-One \(M:1\)](#), [Many-to-Many \(M:N\)](#)

9)

The reviewer entity set is _____ in the diagram given below



(0.5)

[Strong](#), [Weak](#), [Super](#), [Semi strong](#)

10)

_____ is an abstraction through which relationships are treated as higher-level entities

[Aggregation](#), [Specialization](#), [Generalization](#), [Inheritance](#)

(0.5)

DESCRIPTIVE

Answer all the questions.

11)

Suppose a relational schema $R(a,b,c,z)$ and set of functional dependency as following:

$F: \{b \rightarrow a, az \rightarrow bc, c \rightarrow abz\}$. Find the canonical cover F_c .

(4)

12)

Discuss the functionalities of different components of query processor and storage manager .

(3)

13)

Consider the following schema:

Employee(Fname,Mname,Lname,Essn,bdate,address,salary,Dnumber)

Department(Dname, Dnumber, Mgr_ssn,Mgr_start_date)

Dependent(Essn,Dependent_name,gender,Relationship)

Write **Relational algebra expressions** for the following questions:

a) Retrieve the Fname, Lname and address of all employees who work for the 'Research' department.

(3)

b) Find the average salary of employees in each department.

c) Retrieve the details of employees who have more than two dependents.

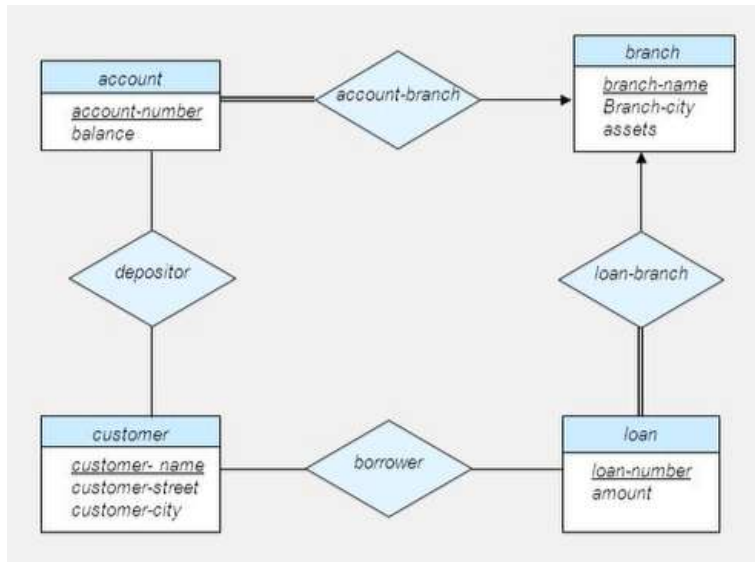
14)

Consider the E-R diagram for a banking enterprise provided below.

(3)

(a) List the cardinality and participation for each relationship in the diagram.

(b) Convert the E-R diagram into a relational schema



- 15) Raju wishes to model an inventory management system with customers, employees and products, that need to be managed. Every customer information stored comes with ID, cname, and a mandatory purchase of atleast one product with purchase date. A customer can buy a maximum of k products and a product can be bought by 1 or m customers. Every product has ID, title and description. Every employee is either a regular employee or a manager, who manages 1 to m products where as a product can be managed exactly one employee. Considering the given scenario, identify primary key, cardinality as low..high with necessary roles of employee, illustrate the scenario with a neat ER model . (3)

- 16) Consider the following schema for Airline Reservation System

- Passengers (PassengerID, Name, ContactInfo)
- Reservations (ReservationID, PassengerID, FlightID, SeatNo, BookingDate)
- Flights (FlightID, Airline, Source, Destination, DepartureTime, ArrivalTime)
- Airlines (AirlineID, Name, Country)

(3)

Using 'with clause' formulate a query to find the FlightId, Name and number of passengers who have made a reservation but have not been assigned a seat for their flight. Display the flight ID, airline name, and the total count of unassigned reservations per flight.

- 17) Using the airline schema in Q16, write SQL query to retrieve the names of passengers, the airline name, the source and destination of their flight, and their seat number for flights traveling from New York to Los Angeles. Display the results sorted by airline name. (2)

- 18) Consider the following schema and write SQL queries for the following:

Library (Libname, city)

book(ISBN, Libname, title, price)

- a) Using an appropriate join operator, list all the library names with the number of books they have. (2)

- b) Rewrite the following query using join operator:

(Select Libname from library) except (select distinct(Libname) from book).

- 19) Describe the ways to create user-defined types in SQL. (2)